# TRCM System Specification

## User Flow Documentation: Claim Engine & TRCM Coding Agent

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# 1. Claim Engine — User Flow

## 1.1 Single Claim Search, Build & Review Flow

**Step 1: Search Encounter or Patient**

- *User Action:* Enter Patient ID (e.g., TRCM-PAT-005) or Encounter ID (e.g., TRCM-ENC-005) into search bar and click **Search**.
- *System Action:* Fetches matching records and displays Encounter Card Details box.

**Step 2: Open Encounter Details**

- *User Action:* Click the **Open** text on the Encounter Card Details box.
- *System Action:* Expands and displays the detailed encounter view.

**Step 3: EDI Configuration & Auto-Lookup**

- *User Action:* Click **EDI Config** button to enter Sender ID, Receiver ID, and NPI value if missing.
- *System Action:* If NPI is missing during build, system alerts to enter correct NPI in EDI Config. Upon entering NPI, the system executes an automated **NPPES Registry lookup** to auto-populate State, ZIP, and Taxonomy Code.

**Step 4: Build Claim**

- *User Action:* Click **Build Claim**.
- *System Action:* Stores claim data in FHIR server and generates the **EDI X12 837P** file for viewing.

**Step 5: Run Claim Review**

- *User Action:* Click **Review** button.
- *System Action:* Sends claim payload to **TriZetto APIs** (Action Edits and PCO) to perform validation checks.

**■■ Technical Note — Intermittent PCO API Exception:**
Occasionally, the TriZetto PCO API may return an HTTP 503 Service Unavailable HTML response instead of valid JSON data: <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN""http://www.w3.org/TR/html4/strict.dtd"> <HTML><HEAD><TITLE>Service Unavailable</TITLE> <META HTTP-EQUIV="Content-Type" Content="text/html; charset=us-ascii"></HEAD> <BODY><h2>Service Unavailable</h2> <hr> HTTP Error 503. The service is unavailable. </BODY></HTML>
*Handling:* The application should catch this HTML error response gracefully, display a user-friendly retry message, and prevent the review interface from crashing.

**Step 6: Handle Validation Edits & Rebuild**

- *User Action:* Correct flagged parameters (e.g., Payer ID: 87726, Trading Partner: 87726, State: NY, Taxonomy Code: 207Q00000X, NPI: 1245319599) and click **Rebuild Claim**.
- *System Action:* Regenerates the claim payload and re-runs validation.

**Step 7: View Review Results**

- *User Action:* Inspect feedback across three dedicated tabs: **Issues & Fixes**, **PCO Prediction**, and **Payer Policy**.

## 1.2 Batch Review Flow

**Step 1: Upload & Submit Batch**

- *User Action:* Upload EDI files (Max 10 files, max 50 claims/file) and click **Review**.
- *System Action:* Processes files asynchronously through TriZetto Action Edits and PCO APIs.

**Step 2: Inspect Claim Details**

- *User Action:* Click **Details** on individual claim cards to view specific validation findings.

**Step 3: Interactive Assistant / Chat**

- *Single Claim Chat:* Click **Ask** on a claim card to launch a chat scoped *strictly* to that individual claim.
- *Batch Chat:* Use the main chat panel to ask questions across the *entire* batch dataset.

### 1.3 Claim Engine Validation Matrix

| Parameter / Feature | Rule / Value | Description |
|---|---|---|
| Search Identifiers | Patient / Encounter ID | e.g., TRCM-PAT-005 or TRCM-ENC-005 |
| Auto-Lookup Registry | NPPES Registry API | Fetches State, Zip, and Taxonomy Code via NPI. |
| Generated Format | EDI X12 837P & FHIR | Stored in FHIR server and exported to EDI 837P. |
| Validation Engine | TriZetto APIs | Action Edits & Predictive Claim Outcomes (PCO). |
| Batch Limits | 10 Files / 50 Claims | Max 10 EDI files per batch, 50 claims per file. |

## 2. TRCM Coding Agent — Demo Flow & Script

### 2.1 Single-Note Coding Flow

**Step 1: Patient Search**

- *User Action:* Enter TRCM-PAT-005 in Patient ID field and click **Search**. *(Alt samples: TRCM-PAT-001 to 004)*.
- *System Action:* Fetches matching SOAP notes.

**Step 2: Load SOAP Note**

- *User Action:* Click **Select** on the SOAP Note card.
- *System Action:* Displays clinical SOAP note contents in the viewer.

**Step 3: Execute Medical Coding**

- *User Action:* Click **Code Note** at the top of the window.
- *System Action:* Coding Agent processes note and outputs results concurrently to: **Coding Results Pane** and **Chat Pane**.

### 2.2 Interactive Chat Pane & Inquiries

**Agent Activity Transparency:** Displays tool calls, vector search arguments, and reference responses in real-time.

**Mirrored Output:** Mirrored presentation of assigned ICD-10 / CPT codes with supporting clinical evidence.

**Case Follow-ups:** User can ask specific questions regarding coding rationale (e.g., *'Why CPT 99214 instead of 99213?'*).

**General Knowledge Queries:** User can ask general medical coding questions (e.g., *'What are the ICD-10 codes for diabetes?'*).

### 2.3 Automated Audit Flow (LLM-as-a-Judge)

**Step 1: Trigger Compliance Audit**

- *User Action:* Click **Audit** (bottom-left of window).
- *System Action:* Invokes Audit Agent to review coding against compliance rules and documentation guidelines.

**Step 2: Review Audit Findings**

- *System Action:* Outputs compliance score, flagged issues, and corrections simultaneously to **Audit Results Pane** and **Chat Pane**.

## 2.4 Coding Agent Architecture & Talking Points

| Agent Component | Core Architecture | Primary Responsibilities |
|---|---|---|
| **Coding Agent** | Dynamic Tool-Using AI Agent | • Reads unstructured SOAP notes.<br>• Executes tool calls against ICD/CPT reference PDFs & text.<br>• Maps diagnoses and procedures to standard medical codes. |
| **Audit Agent** | LLM-as-a-Judge Compliance Engine | • Reviews output of Coding Agent against SOAP note.<br>• Verifies documentation integrity & modifier usage.<br>• Recommends actionable corrections and flags risks. |